Rayleigh's Dissipation Function at Work

Classic Mechanics Term Report

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Summary

1 Introduction

2 Generalized Friction Protential
- Derivation Process
- Application

Introduction

This Report is Based on Thesis: [1]

Rayleigh's dissipation function at work

E. Minguzzi*

Abstract

It is shown that the Rayleigh's dissipation function can be successfully applied in the solution of mechanical problems involving friction non-linear in the velocities. Through the study of surfaces at contact we arrive at a simple integral expression which gives directly the Rayleigh dissipation function in terms of generalized coordinates. In this way the solutions of Lagrangian problems with friction are reduced to often elementary calculations of the kinetic energy, the potential energy and the Rayleigh dissipation function. Some examples of pedagogical interest are given.

Raw Lagrange Equation

As we all know, in case of potential V is independent with velocity of $\dot{q}$ we have Lagrange Equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} + \frac{\partial V}{\partial q^k} = Q_k^{(\mathrm{nc})}. \tag{1}$$

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Superscript, not exponent

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means non-conservative

Rayleigh's Disppative Protential

If the non-conservative force linear correlate with velocities, consider particle a,

$$F_{ai} = -K_{ij} v_a^j, \tag{2}$$

We can verify that

$$Q_k^{(\mathrm{nc})} = -\sum_a \nabla_{\mathbf{v}_a} R \frac{\partial \mathbf{v}_a}{\partial \dot{q}^k} + Q_k^{(\mathrm{nc}')} = -\frac{\partial R}{\partial \dot{q}^k} + Q_k^{(\mathrm{nc}')}. \tag{3}$$

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Lagrange-Rayleigh Equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} + \frac{\partial V}{\partial q^k} = -\frac{\partial R}{\partial \dot{q}^k} + Q_k^{(\mathrm{nc}')}. \tag{5}$$

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Rayleigh Dissipation Protential

Lagrange-Rayleigh Equation

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non-conservative force except Rayleigh Linear Force

What did this paper do?

The Rayleigh Protential is only suitable for linear friction force.

This paper promote Rayleigh Protential to general case when consider contact friction.

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Generalized Friction Protential

Subsection 1

Derivation Process

Generalized Rayleigh Protential

Surfaces contact friction can be write as

$$\mathbf{F} = -N\mu\left(v\right)\hat{\mathbf{v}}, \tag{6}$$

and let $\mathbf{v}_a^{(\mathrm{r})} = \mathbf{v}_a - \mathbf{v}_b$. Here we allow the 2 surfaces move respectively, with consistent the same normal.

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Denote S_1 is the surface we concern and a is a point above it

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Denote S_2 is reference surface
and b is the contact point with a

Generalized Rayleigh Protential

So the tiny friction of point a is

$$\mathbf{F}_a^{(\mathrm{nc})} = -p_a \Delta A \mu \left(v_a^{(\mathrm{r})} \right) \hat{\mathbf{v}}_a^{(\mathrm{r})}. \tag{7}$$

Theorem

If we define the Rayleigh

$$\mathcal{R}(\mathbf{r}, \mathbf{v}) = \Delta A \int_0^{v_a^{(\mathrm{r})}} \mu(u) \, \mathrm{d}u \tag{8}$$

so that we have

$$\mathbf{F}_a^{(\mathrm{nc})} = -\frac{\partial \mathcal{R}}{\partial \mathbf{v}_a} + \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial \mathcal{R}}{\partial \dot{\mathbf{r}}_a} \tag{9}$$

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If we define R as

$$R = \Delta N \sum_a p_a \int_0^{v_a^{(\mathrm{r})}} \mu(v) \, \mathrm{d}v \tag{8}$$

so that indeed

$$\frac{\partial R}{\partial \dot{\mathbf{r}}_a} = \mathbf{F}_a^{(\mathrm{nc})} \tag{9}$$

then the Lagrange equation for the generalized coordinates

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j^{(\mathrm{nc})} \tag{10}$$

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Generalized Rayleigh Protential

Corollary

Let a to be infinitely tiny as well as $\Delta A \to \mathrm{d}A$,

$$R = \int_{S_1} \mathrm{d}A p\left(x\right) \int^{v^{(\mathrm{r})}(x)} \mu \mathrm{d}v. \tag{13}$$

Generalized Rayleigh Protential

Corollary

Let a to be infinitely tiny as well as $\Delta A \to \mathrm{d}A$,

$$R = \int_{S_1} \mathrm{d}Ap\left(x\right) \int^{v^{(\mathrm{r})}(x)} \mu \mathrm{d}v. \tag{13}$$

Subsection 2

Application

A Rotating Disk

Example

A disk of mass m and radius r placed on horizontal surface of friction coefficient μ. At time $t = 0$ disk has angular velocity ω about its center and zero translational velocity. Find out at which time τ it will stop.

Solution

A Rotating Disk

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The Generalized Rayleigh Protential can be written according to formula (13),

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Solution

The Generalized Rayleigh Protential can be written according to formula (13),

$$R = \int_{S_1} \mathrm{d}A \frac{mg}{\pi r^2} \int^{\rho\dot{\varphi}} \mu \mathrm{d}v = \frac{\mu mg}{\pi r^2} \int_{S_1} \mathrm{d}A \int^{\rho\dot{\varphi}} \mu \mathrm{d}v = \frac{2}{3}\mu mgR\dot{\varphi}. \quad (14)$$

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And the Lagrangian $L = T = \frac{1}{4}mr^2\dot{\varphi}^2$. Take them into Lagrange-Rayleigh equation (5),

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}mr^2\dot{\varphi}\right) = -\frac{2}{3}\mu mgR. \tag{15}$$

Thus $\ddot{\varphi} = -\frac{4}{3}\frac{\mu g}{r}$, and

$$\tau = \frac{3\omega R}{4\mu g}.$$

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How about more complex case?

A Rotating Disk

Lemma

(Reye's Hypothesis) If the system exhibits point-by-point identical wear on contact surface, the μ satisfied

$$\mathrm{d}N = \frac{k}{v^{(\mathrm{r})}(x)}\mathrm{d}A, \tag{16}$$

the k is a coefficient satisfing the normalization condition.

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A Rotating Disk

Example

Find out τ under Reye's Hypothesis.

Solution

Using equation (15) we get

A Rotating Disk

Example

Find out τ under Reye's Hypothesis.

Solution

Using formula (13) again,

A Rotating Disk

Example

Find out τ under Reye's Hypothesis.

Solution

Using formula (13) again,

$$\begin{aligned} \tau &= \int_{S_1} \mathrm{d}A\, p(r) \int_0^{r_0} r \mathrm{d}r \\ &= \pi \int_{S_1} \mathrm{d}A\, p(r) \int_0^{r_0} \mathrm{d}r \end{aligned}$$

A Rotating Disk

Example

Find out τ under Reye's Hypothesis.

Solution

Using formula (13) *again,*

$$R = \int_{S_1} \mathrm{d}Ap\left(x\right) \int^{\rho\dot{\varphi}} \mu \mathrm{d}v$$

$$= \mu \int_{S_1} \mathrm{d}Ap\left(x\right) \int^{\rho\dot{\varphi}} \mathrm{d}v$$

A Rotating Disk

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Find out τ under Reye's Hypothesis.

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$$= \mu \int_{S_1} \mathrm{d}Ap\left(x\right) \int^{\rho\dot{\varphi}} \mathrm{d}v$$

$$= \mu \int_{S_1} \mathrm{d}N \rho\dot{\varphi}$$

$$= \mu k \int_{S_1} \frac{\mathrm{d}A}{\rho\dot{\varphi}} \rho\dot{\varphi}, \qquad \left(\int_D \mathrm{d}N = k \int_D \frac{1}{v^{(\mathrm{r})}} \mathrm{d}A = mg \right)$$

$$= \mu m g \left[\int_D \frac{1}{\rho\dot{\varphi}} \rho \mathrm{d}\rho \mathrm{d}\varphi \right]^{-1} \int_D \mathrm{d}A = \frac{1}{2} \mu m g r \dot{\varphi}$$

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$$
\begin{aligned}
&= \mu \int_{S_1} \mathrm{d}N \rho\dot{\varphi} \\
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Reye's Hypothesis

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&= \mu mg \left[\int_D \frac{1}{\rho\dot{\varphi}} \rho \mathrm{d}\rho \mathrm{d}\varphi \right]^{-1} \int_D \mathrm{d}A = \frac{1}{2} \mu mgr\dot{\varphi}
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\tag{17}
$$

Reye's Hypothesis

$$
\begin{aligned}
&= \mu \int_{S_1} \mathrm{d}N \rho\dot{\varphi} \\
&= \mu k \int_{S_1} \frac{\mathrm{d}A}{\rho\dot{\varphi}} \rho\dot{\varphi}, \qquad \left(\int_D \mathrm{d}N = k \int_D \frac{1}{v^{(\mathrm{r})}} \mathrm{d}A = mg \right) \\
&= \mu mg \left[\int_D \frac{1}{\rho\dot{\varphi}} \rho \mathrm{d}\rho \mathrm{d}\varphi \right]^{-1} \int_D \mathrm{d}A = \frac{1}{2} \mu mgr\dot{\varphi}
\end{aligned}
$$

So $\tau = \frac{\omega r}{\mu g}$, using Lagrange-Rayleigh equation.

References I

1. Minguzzi, E. Rayleigh' s dissipation function at work. *European Journal of Physics* **36,** 035014 (2015).

The End